

1096

Brace Expansion 1)

"{a,b}{c,{d,e}}"

ac, a{d,e}, bc, b{d,e}

ac, ad, ae, bc, bd, be

"{{a,z}, a{b,c}, {ab,z}}"

{a,z}, ab, ac, {ab,z}

a, z, ab, ac, ab, z

a, ab, ac, z

February 2011

1 Tuesday

}} } }} }

~~~~~

~~~~~

~~~~~

big problem → small problem

" Recursion

2 Wednesday

```
performUnion() {
```

```
    set <string>
```

```
    temp = doSomething();
```

```
}
```

← This will make  
the union of comma-  
separated entities.

Concatenation :

" b { c } "

↓

" b "

getUnit(c)

↓

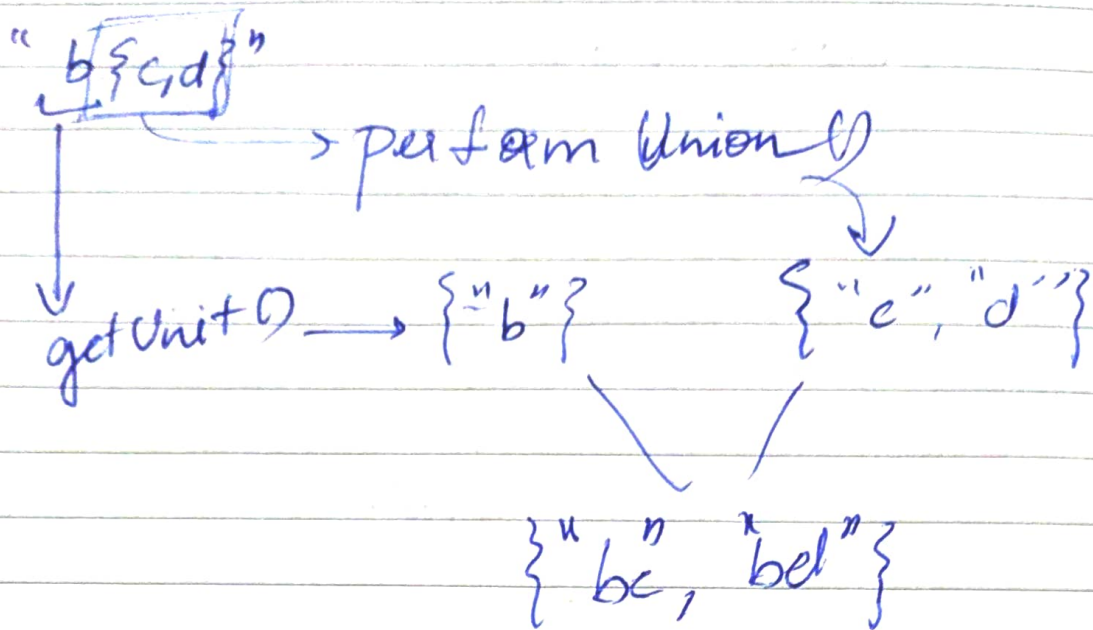
{ " b " }    { " c " }

↓

{ " bc " }

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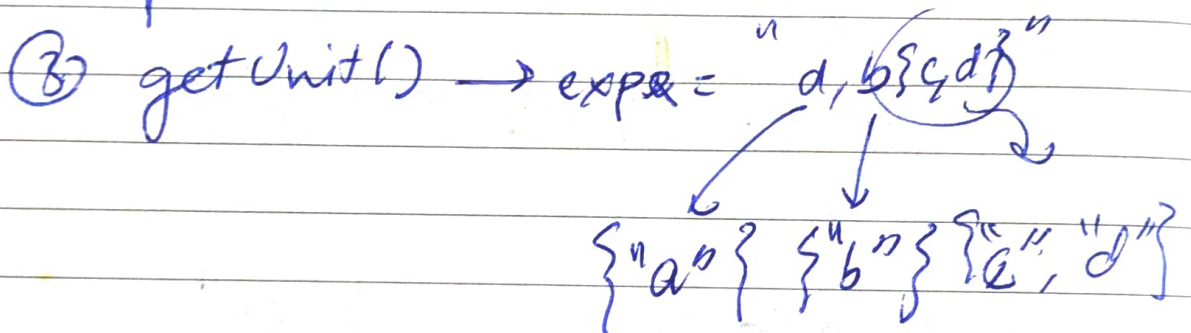


So, three functions:

4 Friday

① performUnion

② performConcatenation



set < string > getUnit()

set < string > result;

if ( expression[index] == '{' )

index++;

result = performUnion();

} else {

result = string(1, expression[index])

}

index++; return result;

}

February 2011

| S  | M  | T  | W  | T  | F  | S  | S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 |

April 2011

20 Wednesday

```
Set<string> performConcat() {
```

```
    Set<string> result;
```

```
    while (idx < n && (expr[idx] == '{' ||
```

```
        isalpha(expr[idx]))) {
```

```
        Set<string> tempright = getUnit();
```

```
        Set<string> concatResult;
```

```
        for (left: result) {
```

```
            for (right: temp) {
```

```
                concat.insert(left + right);
```

```
            }
```

```
        result = concatResult;
```

```
    }  
    return result;
```

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